Heat Pump Proposal

Customer details

Bugs Bunny
39 St. Floras Road, Littlehampton, BN17 6BH
19 York Road

12 Mar 2026
Quote valid for 30 days

Heating R Us
12 Montgomery Crescent
Romford
RM3 7UJ

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1.1 Main Components & System Upgrades

Main components

Heat Pump

Vaillant

Vaillant Arotherm Plus 7kW

Cylinder

Vaillant

200L Unvented

Proposed Flow Temperature 45 °C Nominal Output undefined kW Output at Design Conditions 8.63 kW Heating SCOP at 45°C 3.91

Hot Water Temperature 50 °C Capacity 200 litres Recovery Rate 73 mins Immersion Heater Size undefined kW

1.2 Quotation

Heat Pump and Installation £11,100.00 Total (excl. VAT) £11,100.00 VAT (%)

£0.00

BUS Voucher -£7,500.00 TOTAL PAYABLE £3,600.00

Please note: The contribution from the Boiler Upgrade Scheme (BUS) that has been applied to the price shown is dependent on the BUS voucher being redeemed from Ofgem. A BUS voucher being issued does not guarantee redemption.

2. Home Survey and System Design

There are three main parts of your heating system that need to be sized correctly for your home:

Heat Pump

Hot Water Cylinder

Emitter System (radiators and/or underfloor heating)

To design your system we have calculated how much energy is needed to heat your home and hot water throughout the year. We take into account size, building materials and also your home's location. For example, what material the walls are made of.

We have designed your proposed system for a maximum flow temperature of:

45°C This means that when it is cold outside, this is the hottest your radiators should need to get to keep your house warm.

Heat Pumps Work Best Low and Slow

Heat pumps are more efficient at lower flow temperatures. They use less electricity heating your home when they are on for longer periods at a lower flow temperature than when they are on for shorter periods at a higher flow temperature. This is similar to a car using less fuel to travel the same distance at 50mph than it would driving at 70mph.

This means that with a heat pump your heating may be on for more of the day and your radiators may not get as hot as you are used to.

3. Heat Pump

We have calculated your home's heating requirements to be:

Heating Power Requirement

4.93 kW

How much heat your heating system needs to deliver to keep your home warm in winter.

Annual Heating Requirement 11,163 kWh

This is the amount of heat your home will need over a whole year.

Based on these requirements your proposed heat pump is:

Vaillant

Vaillant Arotherm Plus 7kW

NOMINAL OUTPUT

kW

This is the number that you will usually hear referred to as the ‘size’ of the heat pump. Heat pump sizes are usually quoted at industry standard temperatures of 7°C/35°C. For example, a 10 kW air source heat pump will deliver 10 kW of heat to a home if the outdoor temperature is 7°C and the flow temperature is 35°C.

OUTPUT AT DESIGN CONDITIONS

8.63kW

This is the amount of heat your heat pump will deliver when it is cold outside. For your heat pump to be able to heat your home in the winter without help from any other heat source, this number should be larger than your home’s Heating Power Requirement at the top of the page.

SEASONAL COEFFICIENT OF PERFORMANCE (SCOP)

3.91

This is a calculation of the efficiency of your proposed heating system over a whole year.

Heat Pump Efficiency and SCOPs

The Seasonal Coefficient of Performance (SCOP) is an industry standard way to measure how efficient a heat pump is at producing heat from electricity throughout the year. It's calculated by dividing the total amount of heat delivered by the total amount of electricity consumed over the year.

For example, a heat pump with an SCOP of 3 will produce 3kWh of heat for every 1kWh of electricity used. For fossil fuel based systems like gas or oil boilers, efficiency is less than one with usually 1kWh of fuel producing only between 0.8 – 0.9kWh of heat.

Heat pumps are more efficient than other types heating systems as the heat they provide is harvested from the environment (air or ground) rather than coming directly from an energy source, such as natural gas.

4. Hot Water

We have calculated the required capacity of your hot water cylinder using assumptions for the average hot water use per person and the potential occupancy level of your home based on the number of bedrooms.

Required hot water capacity

120 litres

Vaillant

200L Unvented

Based on these requirements your proposed hot water cylinder is:

CYLINDER CAPACITY

Hot water temperature 50 °C

200litres This is the amount of hot water that your chosen cylinder can hold.

RECOVERY RATE (cylinder reheat time)

73 mins This is how long it would take for your heat pump to replace the hot water in your cylinder once it has been used.

IMMERSION HEATER SIZE

kW The immersion heater is mainly used for the legionella cycle. The immersion can also be used to boost your hot water if you need it but it is less efficient than using the heat pump to heat the hot water.

5. Room by Room Heat Loss

We have completed a room-by-room assessment to work out whether any of your radiators need upgrading as part of your installation. This assessment considers:

• the building materials of your home
• the size of rooms and what they are used for

Below you can see for each room:

• the heat output required to keep the room at the design temperature
• how well the existing radiators meet the heating requirement (if applicable) • whether any replacements or additional radiators are being proposed

LOUNGE (14.91m^2)

DESIGN REQUIREMENTS

Room Temperature 21°C Power

746W

UFH

Power -W Performance 100%

DINING ROOM (8.64m^2)

DESIGN REQUIREMENTS

Room Temperature 21°C Power

546W

BATHROOM (8.94m^2)

DESIGN REQUIREMENTS

Room Temperature 21°C Power

837W

KITCHEN (10.12m^2)

DESIGN REQUIREMENTS

Room Temperature 21°C Power

719W

UFH

Power -W Performance 100%

HALL (7m^2)

DESIGN REQUIREMENTS

Room Temperature 21°C Power

375W

FRONT BEDROOM (16.85m^2)

DESIGN REQUIREMENTS

Room Temperature 21°C Power

583W

UFH

Power -W Performance 100%

BEDROOM 2 (9.09m^2)

DESIGN REQUIREMENTS

Room Temperature 18°C Power

278W

UFH

Power -W Performance 100%

RADIATOR

Power 918W Performance 100%

LANDING (7.5m^2)

DESIGN REQUIREMENTS

Room Temperature 21°C Power

261W

RADIATOR

Power 574W Performance 100%

UFH

Power -W Performance 100%

RADIATOR

Power 574W Performance 100%

UPSTAIRS BATHROOM (9.1m^2)

6. Performance Estimate and Carbon Savings

We have calculated the expected performance of your heat pump system, taking into account it's efficiency at the proposed flow temperature and how much energy your home will require for heating and hot water.

You can also see how this compares to your existing system.

Your home's energy requirements:

Space Heating

11,163 kWh

System comparison:

Existing system

Efficiency 0.87 Fuel required 15,892 kWh Fuel price 7.8 p/kWh Annual CO_2 3,798 kg Annual Cost £1,240

Proposed system

SCOP 3.91 Fuel required 4,225 kWh Fuel price 24.8 p/kWh Annual CO_2 1,229 kg Annual Cost £1,048

Hot Water

2,397 kWh

Switching to a heat pump will cut your CO_2 emissions by 2,569 kg of carbon each year:

Return flight from Heathrow to Honolulu, Hawaii

0

Same as driving

8,314

miles

£1,048

1,300

1,200

1,100

1,000

900

800

700

600

500

400

300

200

100

Cost per year (£)

£1,240

Same as planting 117

trees

Existing System Heat Pump

Heating System Type

7. Sound Assessment

For your air source heat pump installation to be allowed under Permitted Development, the noise level of the heat pump must be at or below 42 decibels (dB) when measured at the nearest habitable room from the proposed heat pump location. If the noise level is louder than this then planning permission will be required.

Calculated noise level

dB

Noise level 41dB

Planning permission is not required for this

Vaillant

Vaillant Arotherm Plus 7kW
Result PASS
installation due to noise level

Please note: there may be other factors that mean planning permission is still required, please confirm with your installer.

Air Source Heat Pumps and Planning Permission

If your installation meets certain conditions, you can install an Air Source Heat Pump under 'Permitted Development'. Permitted Development means you do not have to apply for Planning Permission. To install under Permitted Development, your heat pump needs to meet the following conditions:

• The installation must comply with the Microgeneration Certification Scheme (MCS) standards (or equivalent)
• The outdoor unit must be at least one metre from your property's boundary (or three metres in Wales)
• The outdoor unit must not be larger than $0.6m^3$ (or $1m^3$ in Wales)
• The noise level of the heat pump must be at or below 42 decibels (dB) when measured at the nearest habitable room from the proposed heat pump location

If there is already a heat pump or a wind turbine on site, you will need to apply for Planning Permission. If your property is in a conservation area, further conditions apply, and you should speak to your local planning authority.

8. Next Steps

If you have any questions about this proposal please contact Robert Butler at contracts@suncomfort.org or 01613992549.

9. Additional Quotation Details

9.1 Important notes about this quotation

Boiler Upgrade Scheme (BUS)

If you have indicated that you give consent for us to apply for a Boiler Upgrade Scheme voucher on your behalf, we will make the application as soon as possible. We are required to submit certain information about you and your property to make the application, and you will be contacted by the scheme administrator to provide consent yourself. This should arrive via email.

You will have 14 days to give consent, so we ask that said consent is given in a timely manner for us to begin the installation. If consent is not given within this time, your application may be deemed ‘dormant,’ and we would be unable to progress the application at that time.

Once a voucher is approved, we have 3 months from the date of the voucher to complete the installation for air source heat pumps, and 6 months for ground source, to be able to redeem the funds.

Once the installation is complete, we will then submit information to redeem the voucher funds. If at this point Ofgem requires the installation to be audited, you must provide access to the installation if they request it, otherwise funds may be withheld.

Please note: while an accepted application means technically the funds are available for us to redeem for your installation, until Ofgem accepts the information provided at redemption stage once the installation is complete, it is not guaranteed that these funds will be granted.

Contract terms

We have enclosed a copy of our contract with this quotation. Please read this carefully and contact us without delay if you require further clarification.

It is important that this quotation is read in conjunction with the contract and full performance estimate that accompanies it. If you require clarification on any point please do not hesitate to contact us. We endeavour to present you with as much information as possible to allow you to make an informed decision.

Amendments

Should you wish to make any amendments to this quotation, if the work has not yet begun we will issue you with a new quotation and a full new cancellation period. Once work has begun, if you wish to make a change we may issue a ‘variation to contract’ agreement and further costs may be incurred. Please see your Contract for further details.

If you wish to accept this quotation

If you wish to accept the quotation, please read the Contract carefully. If you are in agreement with our terms and conditions, please sign the accompanying Contract and return it to us with your deposit payment if we have requested one. We will then contact you to arrange the date for the installation.

If you have any questions about this quotation, Contract or any other related issue please do not hesitate to contact us.